Project title: what the project does, in one line

Your name

Computational Macro, University of Mannheim, HWS 2026

1 The model

The problem, the state, the Bellman equation, the parameters and where they come from. The template solves the Aiyagari (1994) economy of session 6: households with CRRA utility ($\gamma = 2$, $\beta = 0.96$) face idiosyncratic income risk on a Rouwenhorst chain with seven states ($\rho = 0.9$, $\sigma_\varepsilon = 0.2$) and a borrowing limit $a \geq 0$; a firm with $Y = K^\alpha L^{1-\alpha}$ ($\alpha = 0.36$, $\delta = 0.08$) rents their savings as capital.

$$v(a, z) = \max_{a' \geq 0} \; u\big(wz + (1 + r)a - a'\big) + \beta \, \mathbb{E}\big[v(a', z') \mid z\big]. \tag{1}$$

2 The method

Which of the methods of the course you use and why this one, grids, tolerances, and the accuracy check you ran. The template uses the endogenous grid method on 200 quadratically spaced asset grid points, Young's method for the stationary distribution, and bisection on the interest rate; the inner tolerances (10^{-8} on the policy, 10^{-12} on the distribution) sit below the outer one (10^{-6} on r). The accuracy check is the Monte Carlo simulation of Table 1: its mean lies within two standard errors of Young's.

3 Results

The figures and tables from `output/`, and what they say. The equilibrium interest rate is $r^* = 2.53$ percent after 16 bisection steps, the capital stock $K^* = 7.57$ against 6.05 under complete markets: precautionary saving raises capital by 25 percent.

Table 1: The stationary equilibrium of the baseline economy

Statistic	Value
Interest rate r^* (percent)	2.53
Capital K^*	7.57
Capital to output K^*/Y^*	3.42
Wage w^*	1.278
Share of households at the borrowing limit	0.026
Capital under complete markets	6.05
Simulated mean assets at r^* ($N = 10{,}000$, 5 seeds)	7.553
Standard error across seeds	0.026

4 What next

What the extension cannot answer yet, and what it would take, in a few sentences.

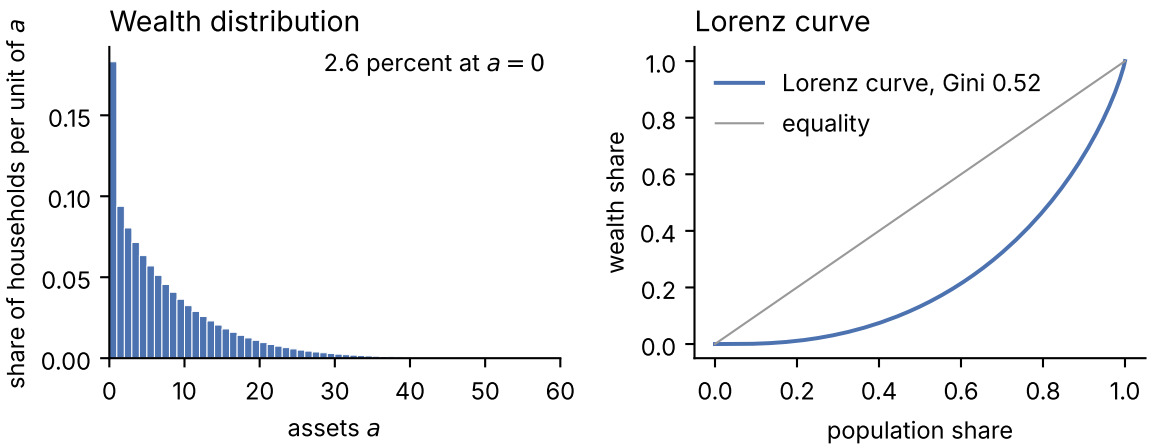


Figure 1: The wealth distribution at r^* (left) and its Lorenz curve (right).